Full Stack development With Mern

1. Introduction

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team Members:

Aahika Sri B

Creation of Flow Designer workflow for standard laptop orders

Anushwa B

Configuration of laptop request form and approval process

Arsha A S

Testing, validation and documentation of the automated procurement flow

2. Project Overview

Purpose: To eliminate manual overhead and reduce fulfillment times in IT device procurement by building an automated end-to-end laptop ordering portal. The system allows employees to select standardized laptop configurations, routes approval requests through manager workflows, and triggers automated provisioning tasks via Flow Designer.

Features:

Interactive Catalog: User-friendly UI for selecting standard laptop tiers (e.g., Developer, Enterprise, Lightweight).

Automated Workflow Orchestration: Real-time workflow triggers for manager approvals and IT fulfillment via ServiceNow Flow Designer integration.

Order Tracking Dashboard: Live status updates for employees showing approval progress, fulfillment stage, and tracking details.

Role-Based Access Control (RBAC): Secure access tiers for Employees, Approvers/Managers, and Procurement Admins.

3. Architecture

Frontend: Built using React.js with functional components, Redux Toolkit for centralized state management, and Tailwind CSS / Material UI for responsive catalog design.

Backend: Built using Node.js and Express.js providing RESTful API routes to process requests, manage authentication, and interface via webhooks/REST APIs with Flow Designer.

Database: MongoDB database schema storing user profiles, catalog item details, order status histories, and system logs.

4. Setup Instructions

The system automates the process of requesting a standard laptop for employees. The employee submits a laptop request through the application, and **ServiceNow Flow Designer** manages manager

approval, inventory checking, laptop allocation, and notifications. This reduces manual work and makes the laptop procurement process faster, easier, and more organized

5. Running the Application

Commands to launch the development servers:

Frontend: Navigate to the client directory and execute npm start.

Backend: Navigate to the server directory and execute npm start (or npm run dev with nodemon).

6. API Documentation

POST /api/v1/auth/login

Request Body: { "email": "user@company.com", "password": "password123" }

Response: 200 OK with JWT Token and User Payload.

GET /api/v1/laptop

Description: Fetches all standard available laptop configurations.

Response: 200 OK with array of laptop catalog objects.

POST /api/v1/order

Description: Creates a new laptop procurement request and triggers Flow Designer workflow.

Request Body: { "laptopId": "651a2b...", "justification": "New Hire - Developer" }

Response: 201 Created with order payload and flow execution ID.

GET /api/v1/orders/:id/status

Description: Fetches real-time status of the request (e.g., Pending Approval, Approved, Provisioning, Shipped).

7. Authentication

Method: JSON Web Tokens (JWT) for session management and standard OAuth2/Basic Authentication for Flow Designer API endpoints.

Flow: Upon logging in, the server generates a signed JWT stored client-side in HTTP-only cookies or local storage. Subsequent requests pass the token in the Authorization: Bearer <TOKEN> header to access protected routes.

8. User Interface & Screenshots

Catalog Selection Interface: Allows employees to compare specs (RAM, Storage, CPU) across standard tiers (Developer, Business, Executive).

Order Tracking Timeline: Displays stage-by-stage progression powered by Flow Designer webhooks.

(Attach system screenshots or animated GIFs here showcasing the catalog page, order submission modal, and status timeline display).

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9. Testing

Testing Strategy: Unit testing for individual React components and Express routes, combined with integration testing for the Flow Designer webhook callbacks.

Tools Used:

Jest & React Testing Library: Frontend unit testing.

Supertest: Express backend API endpoint testing.

Postman: Manual API validation and webhook payload simulation.

10. Known Issues

Webhook Latency: Intermittent 2–5 second delays in updating UI status when Flow Designer processes high volumes of concurrent requests.

Session Expiry on Flow Approval: If a manager takes longer than 24 hours to approve a request, the active user token for status polling may require re-authentication.

11. Future Enhancements

Asset Management Sync: Integration with IT Asset Management (ITAM) databases to auto-assign serial numbers upon delivery.

Slack / Teams Bot Notifications: Automated real-time notifications to managers for quick one-click approvals directly inside Microsoft Teams or Slack.

Predictive Inventory Management: Machine learning models to forecast quarterly laptop demand and auto-reorder stock when standard inventory falls below defined thresholds.